

Gemma Hardware Recommendations & Cost (2025–2026)

A practical buyer's and builder's guide to running Google's **Gemma** open models locally for inference — from per-model memory requirements through hardware options with USD costs and purchase links, a recommended local-first + cloud architecture, security/operations best practices, and a concrete shopping list. Written for the stated goal: **a local Linux box that runs private data locally, calls cloud LLMs only for hard non-private work, stays resilient if any model goes down, and backs up to the cloud** — already using Tailscale + Google Drive, not in healthcare.

Read this first — pricing is abnormal right now. A severe DRAM/GDDR memory shortage in 2026 (foundry capacity monopolized by AI accelerators; GDDR7 scarce) has inflated GPU prices well above MSRP. The RTX 5090 (\$1,999 MSRP) is selling at ~\$2,700–\$4,700; RTX 40-series is discontinued and overpriced on the used market, and DDR5/SSD prices have roughly doubled. Treat every "new" price below as a wide, volatile range and verify live before buying. Datacenter cards (A100/H100/H200) are quoted in \$20k–\$45k+ ranges and are usually *rented*, not bought.

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1. How much memory does each Gemma model need?

LLM token generation is **memory-bound**: the model's weights must fit in VRAM (or Apple unified memory), and generation speed scales with memory *bandwidth*. The single most important number

for "will it run" is the weight size at your chosen quantization, plus KV-cache/runtime overhead.

Quantization in one line: FP16/BF16 = full precision (2 bytes/param); INT8/Q8 $\approx$ half that; INT4/Q4 $\approx$ a quarter — with a modest quality trade-off that is small for Q8 and usually acceptable for Q4 (especially Google's Quantization-Aware-Trained "QAT" int4 checkpoints).

Gemma 3 (1B, 4B, 12B, 27B) — weights only

Model	FP16/BF16	INT8 / Q8	INT4 / Q4 (Google QAT)	Ollama Q4 on-disk
Gemma 3 1B	~2 GB	~1 GB	~0.5 GB	~0.8 GB
Gemma 3 4B	~8 GB	~4 GB	~2.6 GB	~3.3 GB
Gemma 3 12B	~24 GB	~12 GB	~6.6 GB	~8.1 GB
Gemma 3 27B	~54 GB	~27 GB	~14.1 GB	~17 GB

BF16 and int4 figures are from Google's official QAT blog; Q8 is interpolated (~half BF16) since Google publishes official QAT figures only for int4 and BF16. The **27B int4 = 14.1 GB (Google) vs ~17 GB (Ollama Q4_K_M)** gap is expected, not a contradiction: Google quotes pure int4 weights; Ollama's Q4_K_M is a mixed-precision GGUF (some tensors kept at higher bit-width) measured on disk.

Gemma 3n (E2B, E4B — mobile/edge optimized)

"E" = *effective* parameters. Per-Layer Embeddings (PLE) offload embedding tables to CPU RAM, so accelerator memory behaves like a smaller model.

Model	Raw params	BF16 (full)	INT4 / Q4	Official minimum
Gemma 3n E2B	~5B (~2B effective)	~5 GB	~2 GB	2 GB
Gemma 3n E4B	~8B (~4B effective)	~8 GB	~3–4 GB	3 GB

The "2 GB / 3 GB" figures are *minimums* (max quantization, short context, PLE offloaded); BF16 full loads are meaningfully larger (~5 GB / ~8 GB).

Gemma 2 (2B, 9B, 27B) — weights only

No official Google QAT table exists for Gemma 2; figures are weights-only estimates (bytes-per-param) corroborated by Ollama on-disk sizes.

Model	FP16/BF16	INT8 / Q8	INT4 / Q4	Ollama Q4 on-disk
Gemma 2 2B	~5 GB	~2.5 GB	~1.2–1.5 GB	1.6 GB
Gemma 2 9B	~18 GB	~9 GB	~5–6 GB	5.4 GB
Gemma 2 27B	~54 GB	~27 GB	~13.5–16 GB	16 GB

Context / KV-cache overhead (don't forget this)

All weight figures above **exclude the KV cache**, which grows with context length and often decides whether a long-context job fits.

- **Rule of thumb:** add ~15–20% on top of weight size for runtime + activations + short ($\leq 2K$ -token) context. Long 32K–128K contexts can add several GB more, especially on the 12B/27B variants.

- **Gemma 3** uses interleaved local/global attention (5:1 ratio, 1024-token sliding window). Only ~1/6 of layers store K/V for the full 128K window, cutting long-context KV overhead from ~60% (Gemma 1 style) to **under 15%**.
- **Gemma 2** uses a 1:1 local/global ratio with a 4096-token window → higher KV overhead than Gemma 3 at equivalent context.

2. Consumer / prosumer GPUs (NVIDIA + AMD)

The practical dividing line: **24 GB VRAM is the floor for running 27B comfortably at Q4**. 16 GB cards are excellent for the 12B tier but force 27B into degraded Q3 quants. 12 GB handles 4B and 12B-Q4.

GPU	VRAM	~USD price (2026, volatile)	Best Gemma fit
RTX 3060	12 GB	~\$280–\$466 new / ~\$200–\$250 used	4B (any), 12B Q4. Best budget entry ; 2026 relaunch rumored.
RTX 4060 Ti 16GB	16 GB	~\$499 MSRP (often higher)	12B at Q8, or 12B Q4 long-context. Good VRAM/watt.
RTX 4070 / 4070 Ti Super	12 / 16 GB	~\$549–\$700 / ~\$799 MSRP (now ~\$740 used)	12B Q4 (4070); 12B Q8 (Ti Super). Discontinued.
RTX 4080 / 4080 Super	16 GB	\$999 MSRP (inflated now)	12B Q8 fast; 27B only at low (Q3) quant.
RTX 4090	24 GB	~\$1,599 MSRP → ~\$2,300 used / ~\$2,755 new	Classic 27B-Q4 card (~31 tok/s). Discontinued, pricey.
RTX 5070 / 5070 Ti	12 / 16 GB	~\$549 / \$749 MSRP	12B Q4 (5070); 12B Q8 (Ti). Best balanced new buy .

RTX 5080	16 GB	\$999 MSRP, often +45%	12B Q8 fast; 16 GB caps 27B.
RTX 5090	32 GB	\$1,999 MSRP → street ~\$4,200–\$4,700	Best single consumer card: 27B Q4 long-context or near-Q8 (~42 tok/s).
AMD RX 7900 XTX	24 GB	~\$750–\$850 used / ~\$1,100–\$1,400 new	Best value for 27B Q4 — half a 4090's price for the same 24 GB. ROCm 7.x.
AMD RX 9070 XT	16 GB	\$599 MSRP / ~\$689 new	RDNA4, 12B Q8 tier; 16 GB caps 27B. ROCm maturing.

Recommended consumer card by target

Target	Practical minimum	Recommended
Gemma 4B (Q4/Q8)	RTX 3060 12GB (or any 8 GB)	RTX 3060 12GB
Gemma 12B Q4	RTX 3060 12GB	RTX 5070 Ti / 4070 Ti Super (more bandwidth)
Gemma 12B Q8	RTX 4060 Ti 16GB	RTX 5070 Ti / 5080 / RX 9070 XT
Gemma 27B Q4	RTX 4090 24GB or RX 7900 XTX 24GB	RTX 5090 32GB (headroom); RX 7900 XTX for value
Gemma 27B Q8	RTX 5090 32GB (tight)	RTX 5090 32GB, or dual 24 GB / workstation card

3. Apple Silicon Macs (unified memory)

Macs share one memory pool between CPU and GPU. They're a strong, power-efficient option for the 12B–27B tier because you can buy a lot of "VRAM" cheaply — but **memory bandwidth (Pro → Max → Ultra) drives token speed far more than the M-generation number.**

- **Usable as VRAM:** macOS caps GPU memory at ~2/3 of RAM (≤ 36 GB machines) or ~75% (> 36 GB). Raise it with `sudo sysctl iogpu.wired_limit_mb=<MB>` (leave 8–16 GB for the system). This matters mainly for 27B-Q8 / large context on 32–48 GB Macs.
- **Bandwidth by tier:** base M4 ~120 GB/s · M4 Pro ~273 GB/s · M4 Max ~410–546 GB/s · M3 Ultra ~800 GB/s. A Max roughly doubles a Pro's token rate; the Ultra is the fastest single box and the only path to 128–512 GB.
- **Reference speed:** Gemma 27B Q4 $\approx$ **24 tok/s on M3 Ultra** (Ollama; MLX/LM Studio a bit faster). Backend speed: $\text{MLX} \geq \text{LM Studio} \geq \text{llama.cpp Metal} \approx \text{Ollama}$.

Target	Recommended Mac	Chip / Bandwidth	RAM	~USD
4B — entry	MacBook Air M4 13"	M4 / 120 GB/s	16 GB	\$999
4B + light 12B	Mac mini M4	M4 / 120 GB/s	16 GB	\$599 (24 GB ~\$799)
12B sweet spot	Mac mini M4 Pro	M4 Pro / 273 GB/s	24 GB	~\$1,399
12B comfortable	Mac mini M4 Pro	M4 Pro / 273 GB/s	48 GB	~\$1,999
27B Q4 — best laptop	MacBook Pro 14/16" M4 Max	M4 Max / 410–546 GB/s	48 GB	~\$3,200–3,500
27B Q4 — best value desktop	Mac Studio M4 Max	M4 Max / 410–546 GB/s	64 GB	~\$2,500 (\$1,999 base 36 GB)
27B Q8 / long context	MacBook Pro 16" M4 Max	M4 Max / 546 GB/s	64–128 GB	~\$4,159 → \$4,700+
27B fastest + future-proof	Mac Studio M3 Ultra	M3 Ultra / 800 GB/s	96 GB (→512)	\$3,999 → \$14,099 maxed

Memory caps to know: MacBook Air M4 maxes at 32 GB; Mac mini M4 Pro maxes at 48 GB; M4 Max scales to 128 GB; M3 Ultra scales to 512 GB.

4. Workstation / datacenter GPUs (for 27B at full precision or serving)

Gemma 3 27B at **BF16** needs **~54 GB for weights alone** (more with KV cache and batching). A single 48 GB card therefore *cannot* hold full-precision 27B with overhead — you must quantize (INT8/FP8/4-bit) or use two cards. **80 GB+ cards run it comfortably at BF16** with room for long context and batched serving.

GPU	VRAM	~USD purchase (volatile)	Gemma 27B fit / notes
RTX A6000 (Ampere)	48 GB	~\$3,500–\$6,250 (~\$5,200 avg)	Quantize for single-card, or 2× for BF16. Cheapest 48 GB; no FP8.
RTX 6000 Ada	48 GB	~\$6,800–\$8,100	Faster than A6000, FP8 support. Quantize or pair for BF16.
RTX PRO 6000 Blackwell	96 GB GDDR7	~\$11,400–\$13,250 (MSRP raised ~55% in a year)	Best single-card BF16 fit — full weights + big context, newest arch, FP8/FP4.
NVIDIA L4	24 GB	~\$2,000–\$3,000	Too small for 27B even quantized; fine for 1B/4B. Low-power (75W).
NVIDIA L40S	48 GB	~\$7,500	Quantize or 2× for BF16; better serving throughput than A6000, FP8.
NVIDIA A100 40GB	40 GB	~\$10,000–\$15,000 new	Can't hold BF16 27B alone; needs quant or 2×. 80 GB sibling is the better target.
NVIDIA A100 80GB	80 GB	~\$8,000–\$15,000 new; ~\$4k–9k used	Comfortable BF16 fit + KV/batching. Great value vs H100 for serving. No FP8.

NVIDIA H100 80GB	80 GB	PCIe ~\$25k–\$30k; SXM ~\$35k–\$40k	Excellent BF16 serving , high throughput, FP8. Usually rented.
NVIDIA H200 141GB	141 GB	~\$28k–\$45k	Overkill-comfortable: huge context/batch on one card (4.8 TB/s). Rent unless heavy 24/7.

By use case: single full-precision card → RTX PRO 6000 Blackwell 96GB (~\$13k); production serving → A100 80GB (value) or H100 80GB (perf + FP8); max context/batch → H200. Avoid L4 and single A100 40GB for 27B BF16.

5. Cloud GPU rental — the alternative to buying

For bursty, experimental, or low-duty workloads, renting (or just using a per-token API) is almost always cheaper than buying. Specialized providers (Vast.ai, RunPod, Lambda) run **2–5× cheaper** than the AWS/GCP/Azure hyperscalers for the same silicon.

Provider	GPU	~USD/hour (on-demand)
Vast.ai	RTX 4090 24GB	\$0.35–\$0.40
Vast.ai	RTX 5090 32GB	\$0.36–\$0.45 (up to ~\$2.00)
Vast.ai	H100 80GB	\$2.15–\$4.00+
RunPod	RTX 4090 24GB	\$0.34 (Community) – \$0.69 (Secure)
RunPod	L4 24GB	~\$0.39
RunPod	L40S 48GB	~\$0.79
RunPod	A100 80GB	~\$1.49–\$1.99
RunPod	H100 80GB	~\$2.89 (Secure)
Lambda Labs	A100 80GB PCIe	~\$1.99

Lambda Labs	H100 PCIe / SXM	~\$3.29 / ~\$3.99–\$4.29
AWS	A100 80GB (p4d, 8-GPU)	~\$4.10/GPU
AWS	H100 (p5)	~\$6.88/GPU
GCP	A100 80GB (a2)	~\$3.28/GPU
GCP	H100 (a3-high)	~\$3.00–\$3.67/GPU
Azure	H100 (NC H100 v5)	~\$6.98/GPU

For Gemma you rarely need an H100: 1B–12B run happily on an **L4 (\$0.39)**, **RTX 4090 (\$0.35–0.69)**, or **L40S (\$0.79)**; 27B fits a single **A100 80GB (~\$1.50–2.00)**.

Per-token / serverless APIs (no GPU to manage)

Provider	Gemma offering	Price
Google AI Studio (Gemini API)	Gemma	Free (rate-limited; free-tier prompts may train Google products)
OpenRouter	Gemma free + paid variants	\$0 free variants; paid adds ~5.5% fee, \$5 min credit
Together AI	Gemma serverless	~\$0.20 / 1M input, ~\$0.50 / 1M output (Batch up to 50% off)
Google Vertex AI	Self-deployed Gemma endpoint	Pay underlying GPU machine hourly; can autoscale to zero

The cheapest path for most low-to-moderate volume is **Google AI Studio's free Gemma tier**, then OpenRouter/Together AI per-token, with self-hosted GPU only justified at high sustained volume or for privacy/control.

Rent vs. buy — break-even rule of thumb

Card	Buy	Rent	Break-even
RTX 4090	~\$1,800 + system	~\$0.40/hr	~4,500–5,000 GPU-hours (~6–7 mo @ 24/7)
A100 80GB	~\$15,000–18,000	~\$1.50/hr	~10,000+ hours (>1 yr @ 24/7)
H100	~\$30,000+	~\$3/hr	~10,000 hours (+ obsolescence risk)

Buy only for sustained 24/7 production on a fixed model, or for privacy/always-on needs — and factor in power, cooling, depreciation, and fast GPU-generation turnover. Otherwise **rent** (or use the free Gemma API).

6. Recommended setup: local Linux box + cloud backup

This is the recommended architecture for the stated preference — **own a local Linux box for day-to-day inference, with the cloud as failover/burst capacity and for off-site backup**. You get data privacy, no per-hour meter for routine use, and a safety valve for spikes or hardware failure.

Hardware — pick by the largest model you run regularly

Tier	GPU	VRAM	~USD (GPU)	Runs
Budget	RTX 3060 12GB (or used 4070 Ti Super 16GB)	12–16 GB	~\$280–\$740	4B any · 12B Q4/Q8
Sweet spot (recommended)	AMD RX 7900 XTX 24GB or used RTX 3090 24GB	24 GB	~\$750–\$900	27B Q4 with real context
Headroom	RTX 5090 32GB	32 GB	~\$2,000 MSRP (street higher)	27B near-Q8, long context

Pro single-card BF16	RTX PRO 6000 Blackwell 96GB	96 GB	~\$11k–\$13k	27B full precision + serving
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NVIDIA vs AMD on Linux: NVIDIA + CUDA remains the most frictionless path (best driver/runtime support across Ollama, llama.cpp, vLLM). A **used RTX 3090 24GB** (~\$700–\$900) is often the best value-per-VRAM local card and is CUDA-native. AMD's **RX 7900 XTX 24GB** is cheaper still and well-supported by ROCm 7.x for inference, but expect occasional rough edges versus CUDA. If you want zero driver hassle, go NVIDIA; if you want maximum VRAM-per-dollar and don't mind ROCm, go 7900 XTX.

A sensible **recommended build**: Ryzen 7/9 or Core i5/i7 CPU, **64 GB system RAM** (lets you offload layers to CPU and run bigger quants than VRAM alone allows), 1–2 TB NVMe, a 24 GB GPU, and a 750W+ PSU. Budget roughly **\$1,500–\$2,000 all-in** with a used 3090, or **~\$2,500–\$3,000** with an RTX 5090.

Linux software stack

- **Distro:** Ubuntu 24.04 LTS or Debian 13 — best driver/CUDA/ROCm coverage and the path most guides assume.
- **NVIDIA:** install the proprietary driver + CUDA toolkit (`nvidia-driver-xxx`, CUDA 12.x). Verify with `nvidia-smi`.
- **AMD:** install **ROCm 7.x** (check your GPU is on the supported list — gfx1100 for the 7900 XTX) and use ROCm-enabled builds of llama.cpp/Ollama.
- **Inference runtime:**
- **Ollama** — easiest; one command pulls quantized Gemma (`ollama run gemma3:27b`). Great default for a personal box.

- **llama.cpp** — most control over quantization, GPU/CPU offload split, and context; pair with GGUF quant (e.g. Q4_K_M).
- **vLLM** — if you'll serve multiple users/high throughput from the box.
- **LM Studio** — GUI option if you want a desktop app on Linux.
- **Remote access:** expose Ollama's OpenAI-compatible API on your LAN, or tunnel via Tailscale/WireGuard for secure access from elsewhere.

Cloud as backup / burst

Two distinct meanings — set up both:

1. **Compute failover / burst** — when the box is busy, down, or a job needs more VRAM than you own (e.g. 27B at full BF16, or many concurrent users): - **On-demand rental:** RunPod or Vast.ai — RTX 4090 ~\$0.35–0.69/hr, A100 80GB ~\$1.50–2.00/hr. Because Ollama/vLLM expose the same OpenAI-style API locally and in the cloud, you can point your client at either with a config flip. - **Zero-ops fallback:** Google AI Studio's **free Gemma API** or OpenRouter/Together AI per-token — ideal as an emergency backstop with no instance to spin up.
2. **Data / model backup** — back up your models, fine-tunes, prompts, and configs off-site: - Object storage (Backblaze B2 ~\$6/TB-month, Cloudflare R2, or AWS S3) synced with **restic** or **rclone** on a cron/systemd timer. Re-downloadable base weights don't need backup — prioritize your *own* artifacts (LoRAs, datasets, configs, `~/.ollama` customizations).

Net cost picture: a one-time ~\$1,500–\$2,000 local box handles all routine Gemma work at zero marginal cost; cloud rental is reserved for overflow (a few dollars per burst) and backup runs ~\$5–\$10/month for storage. This beats renting 24/7 within ~6–7 months for a 4090-class workload (see break-even table above).

7. Making the architecture HIPAA-compliant

Reality check first. HIPAA compliance is *organizational and legal*, not a setting you switch on. No architecture is "HIPAA compliant" by itself — your *organization* is, once it has signed BAAs, a documented risk analysis, policies, workforce training, and breach-notification procedures in place. What follows is an architecture **designed to support** HIPAA compliance for a system handling **PHI** (Protected Health Information). This is engineering guidance, not legal advice — have counsel/your compliance officer review before going live with real PHI.

The one rule that reshapes the design: BAAs

Under HIPAA, **any third party that can access PHI is a Business Associate and must sign a Business Associate Agreement (BAA)** with you. No BAA → you may not send PHI to that service, period. This forces two corrections to Section 6:

- ■ **OpenRouter is out of any PHI path** — aggregators don't sign BAAs.
- ■ **Free API tiers are out** — Google AI Studio's free Gemma tier, Claude Free/Pro/Max/Team, and OpenAI consumer tiers are **not** BAA-eligible (and some train on your prompts).

Cloud LLMs that DO offer a BAA (verified June 2026 — confirm current terms):

Provider	BAA-eligible surface	Notes
Anthropic (Claude)	First-party Messages API + Enterprise	NOT Console/Workbench or Free/Pro/Max/Team. Covered models require 30-day retention (no zero-retention). BAAs signed after 2025-12-02 cover API + Enterprise together.
OpenAI	API endpoints that are ZDR-eligible	Request via baa@openai.com . Live web-search tool is not eligible. ZDR needs a managed/Enterprise agreement.
Google Cloud Vertex AI	Vertex AI (Gemini, and Gemma via Model Garden)	Sign the Google Cloud BAA at the org level ; use only HIPAA-eligible services; consumer Gemini / AI Studio are not covered.

For the cloud leg, pick **one or two** of these under signed BAAs and route only to them. Your **local Gemma box never needs a BAA** — PHI that stays on hardware you control isn't disclosed to a third party. That's the strongest argument for the local-first design here.

Redaction is defense-in-depth, not legal de-identification

Presidio-style PII stripping (Section 6) is valuable, but **automated redaction does not make data "de-identified" under HIPAA**. True de-identification requires either **Safe Harbor** (removing all 18 specified identifiers *and* having no actual knowledge the residual could re-identify) or **Expert Determination**. Automated detection is best-effort and will occasionally miss an identifier, so:

- **Do not** rely on redaction to send PHI to a non-BAA service.
- **Do** keep redaction as a *minimum-necessary* control on top of a BAA'd channel — send the cloud model the least PHI required for the task.

Technical safeguards on the local Linux box (HIPAA Security Rule)

Because you're self-hosting, *you* own the safeguards a SaaS BAA would otherwise provide:

- **Encryption at rest:** full-disk encryption with **LUKS/dm-crypt** on the box and on every backup target. PHI at rest unencrypted is a finding.
- **Encryption in transit:** TLS 1.2+ for all egress; keep the trusted zone on a private network; remote access only via **WireGuard/Tailscale**, never an open port.
- **Access control & authentication:** unique per-user accounts, **MFA**, role-based access (least privilege), and automatic session logoff. No shared logins.
- **Audit controls:** log every PHI access and every model call. LiteLLM can log request metadata; pair with OS-level `auditd` and ship logs to append-only storage. Retain audit logs (HIPAA expects ~6 years of compliance documentation).
- **Integrity & availability:** checksums/versioning on PHI stores; tested **encrypted backups** with a documented disaster-recovery plan.
- **Physical safeguards:** a local box storing PHI must be **physically secured** (locked room/rack, controlled facility access) — this is a real obligation that cloud users outsource but you now own.

Cloud backup/burst — HIPAA version

Section 6's cloud backup must also sit under BAAs:

- **Backup storage:** use a **HIPAA-eligible, BAA-covered** object store — **AWS S3, Google Cloud Storage, or Azure Blob** (all sign BAAs). **Backblaze B2 does not sign a BAA**, so it's out for PHI backups. Encrypt client-side with restic/rclone regardless.
- **Compute burst:** rent only from a provider that will sign a BAA and lets you run in an isolated, encrypted environment (e.g. a dedicated GPU instance in your BAA'd cloud account). Generic Vast.ai/RunPod marketplace hosts are **not** appropriate for PHI without a BAA and tenancy guarantees — keep PHI workloads on your local box or a BAA'd cloud VM, and reserve cheap marketplace GPUs for non-PHI/de-identified work only.

Administrative requirements (don't skip these)

Architecture alone won't pass an audit. You also need: a documented **risk analysis**, written **policies & procedures**, **workforce training**, a **breach-notification** process, signed BAAs with every vendor, and (if you're a Business Associate yourself) the obligations that flow down from your covered entity.

Net effect on the design

The local-first architecture is *already* the HIPAA-friendly shape: keep PHI on the encrypted local box by default; when a frontier model is genuinely needed, send the **minimum necessary, redacted**, over TLS, **only** to a **BAA-covered** endpoint (Claude Messages API / OpenAI ZDR-eligible / Vertex AI); log everything; back up encrypted to a BAA'd store. OpenRouter, free tiers, and marketplace GPU hosts move to the "non-PHI / de-identified only" lane.

8. Verifying your current stack (Tailscale + Google Drive)

You're already running **Tailscale** (secure transport) and **Google Drive** (cloud storage/backup). Both can meet the Section 7 security bar — the task is verifying they're *configured* to, not just present. Healthcare/HIPAA is not the driver here, so BAAs are optional; the bar is the security posture: encryption, least-privilege access, enforced MFA, audit logging, and keeping private info private.

Prerequisite — confirm this is Google Workspace, not consumer Drive. A custom domain (@cwmi.us) implies Workspace. Only Workspace exposes the admin console, audit logs, and DLP needed to *verify* anything below. If it's a consumer Gmail Drive, you cannot verify to this standard — migrate to Workspace first.

Tailscale checklist (secure transport layer)

- ☐ **Least-privilege ACLs** — Admin console → *Access Controls*. Default-deny; only specific users/devices reach the local box, only on needed ports. Not the default allow-all ("*" : ["*"]).
- ☐ **Device approval ON** — *Settings* → *Device management*. New machines need admin sign-off before joining the tailnet.
- ☐ **Key expiry enabled** — *Machines* list. No device shows "Key expiry disabled"; keys rotate on schedule.
- ☐ **MFA enforced via SSO** — identity provider (Google/Okta). No personal-account backdoors into the tailnet.
- ☐ **No public exposure** — Tailscale **Funnel off** unless deliberately publishing a service; nodes reachable only inside the tailnet.

- [] **ACL tests** — add a `tests` block to the policy file asserting who *can* and *cannot* reach the box, so the config can't silently regress.
- [] **Tailnet lock** (optional, high value) — *Settings* → *Tailnet lock*. Cryptographically blocks rogue node injection.
- [] **Audit logging** — configuration + network-flow logs enabled and shipped somewhere you review.

Local sanity checks: `tailscale status` (who's on the tailnet) and `tailscale ping <box>` (confirm a direct, encrypted path). Transport is WireGuard end-to-end by default — the work is the *access policy*, not the crypto.

Google Drive checklist (cloud storage / backup leg)

First decide the data question: in the Section 6/7 design, **private info should stay local**. So either (a) only non-sensitive / already-sanitized data lives in Drive, or (b) sensitive data is there but **client-side encrypted** so Google holds only ciphertext.

- [] **Workspace + DPA** — confirm it's Workspace (not consumer) and Google's Data Processing Amendment is accepted. (Workspace can also sign a BAA later if healthcare ever returns.)
- [] **External sharing locked down** — *Apps* → *Drive* → *Sharing settings*. Restricted or warn-on-external; default link sharing = "**restricted**," not "anyone with the link."
- [] **2-Step Verification enforced** — *Security* → *Authentication*. Mandatory for all users, not optional.

- [] **Audit logging** — *Security → Investigation tool / Drive log events*. You can see who viewed/shared/downloaded what; alert on external shares.
- [] **DLP rules** — *Security → Data protection*. Block/warn when content matching private-data patterns is shared externally.
- [] **Sharing audit** — run an Investigation-tool query for files shared publicly/externally; remediate anything unexpected.
- [] **Third-party app access** — *Security → API controls*. OAuth apps allowlisted; block unverified apps from Drive scopes.
- [] **Client-side encryption** — *Security → Client-side encryption*. Enable for sensitive folders if private data must live in Drive; Google then holds only ciphertext.
- [] **Encryption baseline** — Google encrypts at rest + in transit by default; fine for non-sensitive data, but **not sufficient alone** for your most private data without CSE.

Highest-value checks

Across both tools, the two checks that surface the most real-world exposure: **(1) enforce MFA** (Tailscale SSO + Workspace 2SV), and **(2) run the Drive external-sharing audit**. Do those first.

Bottom line

- **Tailscale** is the right transport and easy to pass — tighten ACLs to least-privilege, enable device approval + key expiry, enforce MFA via SSO, and add ACL tests.
 - **Google Drive** works as the cloud-backup/collaboration leg for **non-sensitive or client-side-encrypted data**. Decide: keep genuinely private info *off* Drive (local-only, per the architecture), or turn on **CSE + DLP + restricted external sharing** so Drive holds only data Google can't read and you can audit access.
-

9. Operational best practices: reliability, resilience, security, results & backup

The design principle that ties this all together: **the local box is the floor**. Every cloud dependency is an enhancement that can fail without taking you to zero. Build so that if every external LLM is down, you still operate locally — degraded in quality, never offline.

Reliability (keep the local box healthy)

- **Run the runtime as a managed service.** Put Ollama/vLLM under **systemd** with `Restart=always` so a crash self-heals. Add a watchdog that pings the model endpoint and restarts on failure.
- **Keep models warm.** Use keep-alive (e.g. Ollama `OLLAMA_KEEP_ALIVE`) so the weights stay resident and you avoid multi-second cold starts.
- **Monitor the box.** Track GPU temp/VRAM, OOM-kills, disk, and service uptime (Prometheus + node/DCGM exporters, or a simple cron health check). Alert before you run out of VRAM, not after.

- **Timeouts, retries, queueing.** Every model call gets a timeout + bounded retry with backoff; serialize heavy jobs through a queue so concurrent requests don't OOM the GPU.
- **Pin everything offline.** Pre-download and pin model weights + quant by digest so a registry outage never blocks startup. The box must boot and serve with **no network**.

Resilience / failover (when an LLM goes down)

- **Tiered fallback chain** through the **LiteLLM gateway**: primary cloud → secondary cloud → local Gemma 27B → local Gemma 4B. The chain *ends* on local, so there's always an answer.
- **Health-aware routing + circuit breakers.** The gateway marks a failing provider unhealthy and stops hammering it; it retries that provider later instead of blocking every request.
- **Local-only "lifeboat" mode.** A single flag (or automatic trigger when all cloud backends are unhealthy) routes 100% locally. Test it deliberately by pulling the network.
- **Response caching.** A semantic/result cache means repeated or recent queries keep working through an outage and cut load. Cache lives on the local box.
- **Graceful degradation, signalled.** When you fall back to a smaller/local model, **tag the response as "degraded mode"** so downstream users/systems know the quality tier changed.
- **Capacity headroom.** Keep enough free VRAM that the local fallback can actually load when cloud traffic suddenly lands on it — don't size the box to exactly the primary model.

Security (recap, enforced continuously)

- **Encryption everywhere:** LUKS at rest on box + backups; TLS 1.2+ and **Tailscale/WireGuard** in transit; no open inbound ports.
- **Access:** enforced MFA/SSO, least-privilege accounts, auto-logout; secrets (API keys) in a vault or env file with strict perms — never in code or git.
- **Egress boundary:** the trusted zone reaches the internet **only** through the gateway, which applies minimum-necessary redaction before any cloud call.
- **Patch & rotate:** keep OS/drivers/runtime patched; rotate API keys and Tailscale keys on a schedule.
- **Audit:** log every model call + PHI/private-data access to append-only storage; review periodically (ties to Section 8).

Results / quality (don't let reliability cost correctness)

- **Pin model + quant versions.** Don't silently auto-upgrade; a model swap can change behavior. Promote new versions deliberately.
- **Golden eval set.** Keep a small regression suite of representative prompts+expected outputs; run it whenever you change model, quant, or prompt to catch quality drops — *especially* before trusting a fallback model.

- **Validate outputs.** Use structured/JSON mode + schema validation; reject or retry malformed results rather than passing them downstream.
- **Quality-aware routing.** Hard tasks → strongest available model; add a critique/verify pass for high-stakes outputs; keep a human in the loop where it matters.
- **Version prompts in git** alongside the gateway and infra config.

Backup protocols

- **3-2-1 rule:** ≥ 3 copies, on 2 media types, with 1 off-site. Off-site = your **Google Drive** (client-side encrypted) and/or S3/GCS.
- **Back up what you can't re-download.** Base Gemma weights are re-pullable (just record the version+digest); prioritize your **own artifacts**: fine-tunes/LoRAs, datasets, prompts, gateway config, **Tailscale ACL policy**, vector DB/embeddings, and data stores.
- **Encrypt client-side, then sync.** `restic` or `rclone crypt` → Drive/S3 so the cloud holds only ciphertext. Automate via systemd timer/cron; keep **versioned, append-only** snapshots to survive ransomware/accidental deletion.
- **Config-as-code.** Keep box setup (compose/systemd units, LiteLLM config, Tailscale ACLs, prompts) in **git** so the entire environment is reproducible from scratch.
- **Test restores.** A backup you've never restored is a hope, not a backup. Schedule periodic restore drills.

- **Write a DR runbook** with target **RPO/RTO** (how much data you can lose / how fast you must be back) and the exact steps to rebuild the box from backups.

One-line summary

Local Gemma is the always-on floor; LiteLLM gives tiered, health-checked failover ending locally; everything private is encrypted in transit and at rest with enforced MFA and an audited egress boundary; quality is protected by pinned versions + a golden eval set; and a 3-2-1, client-side-encrypted, config-as-code, restore-tested backup regime lets you rebuild the whole box on demand.

10. The optimized plan (recommended build)

This is the single opinionated blueprint, tuned to the stated constraints: **local-first for private data, cloud LLMs for the hard parts, resilient to any model going down, Linux, not healthcare, already on Tailscale + Google Drive**. Everything below is a concrete pick, not a menu.

Hardware — one box

Part	Pick	Why
GPU	RTX 5090 32GB (primary) · <i>RTX 3090 24GB used (~\$800) = value pick</i>	32 GB runs Gemma 3 27B Q4 and keeps a small fallback model warm at the same time — that headroom <i>is</i> the resilience.
CPU	Ryzen 9 / Core i7	Handles redaction + CPU-offload of extra layers.

RAM	64 GB	Lets you offload layers / run bigger quants than VRAM alone.
Storage	2 TB NVMe, LUKS-encrypted	Encryption at rest; room for multiple model weights.
OS	Ubuntu 24.04 LTS	Best NVIDIA/CUDA support; what the guides assume.

All-in: ~\$2,500–3,000 (RTX 5090) or ~\$1,500–2,000 (used RTX 3090).

Software stack — one of each

- **Runtime: Ollama**, run under **systemd** (Restart=always, OLLAMA_KEEP_ALIVE). (*gemma.cpp is the Google-first-party alternative; Ollama wins on ease + speed.*)
- **Models, both kept warm:**
- **Gemma 3 27B (Q4)** — primary local model for private data.
- **Gemma 3 4B** — triage/routing, redaction helper, and fast last-resort fallback.
- **Gateway: LiteLLM proxy** — one OpenAI-compatible endpoint, health-checked tiered failover.
- **Privacy boundary: Microsoft Presidio** — minimum-necessary redaction before any cloud call; default-route private data to local only.
- **Access: Tailscale** (already in place) — hardened ACLs, enforced MFA/SSO, no open inbound ports.

- **Backup: restic** (client-side encrypted) → **Google Drive** (already in place) + a second target (S3/GCS) for 3-2-1.

The one failover chain (drives resilience)

```
complex/non-private task → Claude (API) → Gemini (Vertex) → local Gemma 27B → local Gemma 4B
private/sensitive task   → local Gemma 27B → local Gemma 4B           (never leaves the box)
```

The chain **always ends locally**, so no external outage takes you to zero. Cloud models are used only for hard, non-private (or redacted) work. Since you're not in healthcare, BAAs are optional — but still prefer API tiers that **don't train on your data** and avoid free tiers that do.

Rollout — phased for fastest value, lowest risk

1. **Phase 0 — Harden what you already run** (*do first, cheap*): Tailscale ACLs to least-privilege + device approval + enforced MFA; run the **Google Drive external-sharing audit**, restrict external sharing, enable CSE for sensitive folders. (Section 8.)
2. **Phase 1 — Stand up the box**: Ubuntu 24.04 + LUKS; NVIDIA driver + CUDA; Ollama under systemd; pull Gemma 27B + 4B; expose only over Tailscale. *You now have a working private local LLM.*
3. **Phase 2 — Gateway + failover**: LiteLLM with the fallback chain above + result cache; **test lifeboat mode by pulling the network** and confirming local still answers.
4. **Phase 3 — Privacy boundary**: Presidio redaction + default-local routing; lock egress so the trusted zone reaches the internet only through the gateway.

5. **Phase 4 — Quality + ops:** pin model/quant versions; build a small **golden eval set**; add monitoring/alerts (VRAM, OOM, uptime).
6. **Phase 5 — Backup + DR:** automate restic → Drive/S3 (versioned, append-only); put all config (LiteLLM, systemd units, Tailscale ACLs, prompts) in **git**; run a **restore drill**; write the DR runbook with RPO/RTO.

Running cost after build

- **Routine work:** ~\$0 marginal (runs locally).
- **Cloud burst/complex tasks:** a few dollars/month at low volume (per-token APIs).
- **Backup:** ~\$0 extra on existing Google Drive; ~\$5–10/mo if adding S3/GCS.

Why this is the optimum

It maximizes local/private operation, makes the cloud purely additive (so reliability never depends on it), reuses your existing Tailscale + Google Drive instead of new tooling, and stays a single maintainable box. Scale up later by swapping the GPU (toward RTX PRO 6000 / Mac Studio M3 Ultra) or adding a BAA'd cloud backend — no redesign required.

11. Shopping list — real boxes, pricing & purchase links (June 2026)

Price-shock warning. A 2026 AI-driven memory shortage has roughly **doubled DDR5/SSD prices** and pushed the **RTX 5090 35–100%+ over its \$1,999 MSRP**. Analysts expect elevated prices through 2027–2028. Major retailer pages (Newegg, Best Buy, eBay, Amazon, builder sites) block automated fetching, so every figure below comes from search snapshots and price trackers, **not live page reads** — **confirm price and stock in a browser before buying**. All three 24–32 GB tiers run Gemma 3 27B at Q4 (~16–17 GB) comfortably.

Tier A — Premium: RTX 5090 32GB (the Section 10 primary pick)

32 GB runs Gemma 27B Q4 *and* keeps a small model warm beside it.

GPU alone (cheapest first): | Card | Retailer | ~Price | Link | |-----|-----|-----|-----| | Gigabyte RTX 5090 Aorus Master 32G | Best Buy | ~\$2,700 | [bestbuy.com](https://www.bestbuy.com) | | ASUS TUF Gaming RTX 5090 32GB | Newegg | ~\$2,910 | [newegg.com](https://www.newegg.com) | | Gigabyte Gaming OC / MSI Trio OC | Newegg | ~\$3,300 | [Newegg 5090 list](#) | | MSI Suprim Liquid SOC 32GB (premium) | Newegg | ~\$3,500 | [newegg.com](https://www.newegg.com) |

Prebuilt desktops (64 GB RAM): | System | CPU / RAM / Storage | ~Price | Link | |-----|-----|-----|-----| | **iBUYPower RDY Y70 B05** ■ best value, in stock | Core Ultra 9 285K / 64GB / 4TB | ~\$4,799 | [ibuypower.com](https://www.ibuypower.com) | | NZXT Player Three Prime | Ryzen 7 9800X3D / 64GB / 4TB | ~\$4,999 | [nzxt.com](https://www.nzxt.com) | | Skytech Legacy 4 | Ryzen 9 9950X3D / 64GB / 4TB | ~\$4,999–5,799 | [skytechgaming.com](https://www.skytechgaming.com) | | CyberPowerPC Gamer Supreme | Ryzen 9 9950X / 64GB / 4TB | ~\$5,100 | [bestbuy.com](https://www.bestbuy.com) |

Linux note: the 5090 (Blackwell) needs NVIDIA driver 570+ and CUDA 12.8+ — all of Ollama/llama.cpp/vLLM support it.

Tier B — Value: used RTX 3090 24GB (best \$/VRAM, the value pick)

The consensus value champion: ~87 tok/s class on 27B Q4, ~20% slower than a 4090 at a third of the price, CUDA-native.

Option	~Price	Link
Used RTX 3090 24GB (EVGA/FE preferred)	~\$800–1,050	eBay listings · price tracker · r/hardwareswap
AMD RX 7900 XTX 24GB (new, warranty)	~\$1,100–1,340 new / ~\$790–900 used	Sapphire Pulse @ Newegg · tracker
RTX 4090 24GB (skip — poor value now)	~\$2,200–2,470 used	eBay · tracker

DIY build around a used 3090 (≈ \$2,635–2,770 total): | Part | Pick | ~Price | |-----|-----|-----| | GPU | Used RTX 3090 24GB | ~\$950 | | CPU | Ryzen 7 9700X | ~\$300 | | Motherboard | B650/B650E (AM5) | ~\$170 | | RAM | 64GB DDR5-5600 (*shortage-inflated*) | ~\$600–710 | | SSD | 2TB Gen4 NVMe (990 Pro / SN850X) | ~\$345–370 | | PSU | Seasonic Focus GX-1000 | ~\$140 | | Case + air cooler | Fractal/Phanteks + Peerless Assassin | ~\$130 |

Build live on [PCPartPicker](#) to pull current prices. RAM is ~■ of the cost right now thanks to the shortage; 32 GB (~\$375) also works since the model lives in VRAM. AMD note: on the 7900 XTX, llama.cpp's **Vulkan** backend currently beats ROCm — viable but more setup friction than CUDA.

Tier C — Turnkey Linux AI workstations (Ubuntu/Pop!_OS preinstalled, AI-literate support)

Pay a premium for a tested, driver-ready, warrantied box from a vendor who knows CUDA-on-Linux.

Vendor / model	GPU	Base price	As-configured for 27B	Ubuntu	Link

BIZON V3000 G4 ■ lowest-cost turnkey single-GPU	up to RTX 5090 32GB	\$3,191	~\$3.5k–4.5k w/ 5090	■ preinstalled + AI stack	bizon-tech.com
System76 Thelio Mira	up to RTX 5090 32GB	\$1,699	~\$8.5k–9.5k w/ 5090 (<i>5090 upgrade alone +\$5,299</i>)	■ Pop!_OS/Ubuntu	system76.com
Puget Single-GPU AI tower (buy-409)	up to RTX 5090 32GB	quote	~\$5k–8k (est.)	■ HW supported; not the OS	pugetsystems.com
Maingear MG-1 / Pro WS	RTX 5090 32GB	quote	~\$6k–8.5k (est.)	■ Ubuntu/dual-boot; explicit AI marketing	maingear.com
System76 Thelio Major	RTX PRO 6000 96GB	\$6,999	~\$16k–22k (FP16 27B + headroom)	■	system76.com
Exxact Valence (VWS-135223847)	RTX PRO 6000 96GB	\$6,215	low-to-mid five figures	■ + CUDA-X stack	configurator
Puget Multi-GPU Threadripper PRO (buy-570)	RTX PRO 6000 96GB	quote	~\$13k–18k+	■ HW supported; not the OS	pugetsystems.com

BIZON and System76 ship Ubuntu + a preinstalled AI stack (CUDA, Ollama, vLLM, llama.cpp, PyTorch) — "power on and run Gemma in minutes." **Maingear** explicitly offers Ubuntu/dual-boot and markets local-LLM workstations (its Pro AI line). **Puget** preinstalls Ubuntu and validates the AI stack but warns it does **not** support the Linux OS itself (hardware/labor only). **Origin PC** is a weaker Linux fit — Windows-default; you'd order no-OS (~\$129 off) and self-install (configurator). **Lambda is no longer an option** — it exited hardware in Aug 2025 (its free [Lambda Stack](https://configurator) installer still works on any Ubuntu box). *Turnkey vendor sites all blocked automated fetching, so as-configured prices are estimates — get a live quote from the configurator.*

Which to buy (mapped to the Section 10 plan)

- **Best overall (premium): iBUYPOWER RDY Y70 B05 (~\$4,799)** — RTX 5090 32GB, 64 GB, 4 TB, in stock, cheapest turnkey 5090; wipe and install Ubuntu. Or **DIY 5090 build ~\$3,800–4,400** to save ~\$700–1,200.
- **Best value: DIY used-RTX-3090 build (~\$2,650–2,800)** — runs the exact same Gemma 27B Q4 workload; spend the savings on a second backup drive.
- **Best zero-hassle Linux: BIZON V3000 G4 w/ single RTX 5090 (~\$3.5k–4.5k)** — Ubuntu + AI stack preinstalled, AI-literate support, up to 5-yr warranty.
- **Want FP16 27B / room to grow to 70B+:** System76 Thelio Major or Exxact Valence with RTX PRO 6000 96GB (low-to-mid five figures).

12. Upgradeability & future-proofing

All three recommended boxes can be upgraded — but to **different ceilings**. The parts you can swap anytime are the GPU, RAM (to the board's max), and storage. The part you *can't* change later is the **motherboard/chassis platform** — how many full-speed GPU slots, PCIe lanes, and RAM slots you get is fixed at purchase. So the real future-proofing decision is made on day one: single-GPU forever, or room for a second GPU later.

Option	GPU swap	Add 2nd GPU	RAM ceiling	Notes
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iBUYPOWER RDY Y70 (prebuilt 5090)	■ easy	■■ usually needs new PSU + often new board	128–192 GB (board max)	Standard ATX parts; consumer board = one full-speed GPU slot. Check PSU wattage + case clearance before a bigger card.
DIY 3090 build	■ easiest	■■ consumer AM5 limits to one full-bandwidth slot	128–192 GB	Most flexible by design. Buy 1000W PSU + roomy case up front so a GPU swap is drop-in.
BIZON V3000 G4	■	■ factory 2× RTX 5090 capable	up to 192 GB DDR5	Built for upgrades; vendor-supported without voiding warranty. The headroom is part of the turnkey premium.
Threadripper/Xeon WS (Puget buy-570, System76 Thelio Major, Exxact)	■	■ 2–4 GPUs (lots of PCIe lanes)	256 GB–2 TB ECC	The real multi-GPU / scale-to-70B+ path. Costs five figures.

Decision rule

- **Only ever one big GPU?** Any of the three picks is fine — choose on price. A single 24–32 GB card runs Gemma 27B Q4 indefinitely.
- **Might run 70B+ on two GPUs someday?** Buy multi-GPU-capable hardware **now** — the **BIZON V3000 (2× 5090)** or a Threadripper/Xeon workstation. Retrofitting a second GPU into a consumer prebuilt usually means a new PSU and often a new motherboard — effectively a rebuild.

Upgrade-timing notes

- **Buy the RAM you need at purchase.** The 2026 memory shortage makes adding RAM later unusually expensive — there's no cheap "add it next year" path right now.
 - **GPU is the highest-leverage upgrade** (more VRAM → bigger model / longer context / higher quant). Size the PSU and case for the *next* GPU, not just the current one.
 - **Architecture doesn't change when you upgrade hardware.** The Section 10 stack (Ollama + LiteLLM + Tailscale + Drive) is unaffected by a GPU/RAM swap — and you can always add a BAA'd or rented cloud backend for burst capacity instead of buying more silicon.
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TL;DR recommendations

For your stated goal (local-first, resilient, Linux, Tailscale + Drive): build the Section 10 plan around one of three boxes from Section 11 — ■ **DIY used-RTX-3090 build (~\$2,650–2,800)** for best value, **iBUYPOWER RDY Y70 B05 (~\$4,799)** for a turnkey 5090 you wipe to Ubuntu, or **BIZON V3000 G4 (~\$3.5k–4.5k)** for zero-hassle Linux with Ubuntu + AI stack preinstalled and room to add a 2nd GPU. Run Gemma 3 27B (Q4) + 4B under Ollama, front it with LiteLLM failover ending locally, back up encrypted to your existing Google Drive.

General guidance by use case:

- **Just want to try Gemma cheaply:** use **Google AI Studio's free Gemma API**, or rent an **RTX 4090** at ~\$0.40/hr.

- **Cheapest local box (4B / 12B-Q4): RTX 3060 12GB** (~\$280–\$466) in a PC, or a **Mac mini M4 16 GB** (~\$599).
 - **12B daily driver: RTX 5070 Ti 16GB** (~\$749 MSRP) or **Mac mini M4 Pro 24–48 GB** (~\$1,399–1,999).
 - **27B at Q4, best value: AMD RX 7900 XTX 24GB** (~\$750–\$850 used) or **Mac Studio M4 Max 64 GB** (~\$2,500).
 - **27B with headroom / near-Q8, single box: RTX 5090 32GB** (premium right now) or **Mac Studio M3 Ultra 96 GB+** (~\$3,999+).
 - **27B at full BF16 / serving many users: rent an A100/H100 80GB**, or buy an **RTX PRO 6000 Blackwell 96GB** (~\$13k) for a single full-precision card.
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Compiled June 2026. GPU and cloud pricing is volatile — especially during the current memory shortage — so confirm live figures before purchasing. A few source pages returned HTTP 403 to automated fetching; figures from those rest on search-result extracts plus standard bytes-per-parameter math, and are flagged as approximate throughout.